

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

<i>Criteria</i>	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	<i>Total(10)</i>
<i>Weightage</i>	20%	30%	20%	10%	20%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I J. Ramesh (192311230)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ramesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID Products Purchased

O1 Mobile, Earphones

O2 Mobile, Charger

O3 Earphones, Charger

O4 Mobile, Earphones, Charger O5

Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration.

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. (BL3)

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI

S4 Python, Database, AI

S5 Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. (BL4)

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	Treatments Received
------------	---------------------

P1	Healthcare, Cardiology, ECG
----	-----------------------------

P2	Healthcare, Orthopedics
----	-------------------------

P3	Healthcare, Cardiology
----	------------------------

P4	Healthcare, Cardiology, ECG
----	-----------------------------

P5	Healthcare, Orthopedics
----	-------------------------

Question:

Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. Apriori Algorithm for Online Shopping Transactions

The given dataset contains five online shopping transactions involving **Mobile, Earphones, and Charger**. The task is to design pseudocode for Apriori and show candidate generation and pruning.

Given Dataset

Order ID Products

O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Assumption

Since the PDF does not specify a numerical minimum-support threshold, for illustration we use:

$$\begin{aligned} \text{Minimum Support} &= 60\% \\ \text{Total transactions:} & \\ N &= 5 \\ \text{Therefore,} & \\ \text{Minimum Support Count} &= 0.60 \times 5 = 3 \end{aligned}$$

Step 1: Generate 1-Itemsets

Count each product:

Item	Support Count
------	---------------

Mobile	4
--------	---

Earphones	3
-----------	---

Charger	3
---------	---

All three satisfy the minimum support count of 3.

Therefore,

$$L_1 = \{Mobile, Earphones, Charger\}$$

Step 2: Generate Candidate 2-Itemsets

Possible candidates are:

$$C_2 = \{\{Mobile, Earphones\}, \{Mobile, Charger\}, \{Earphones, Charger\}\}$$

Their support counts are:

$$Support(Mobile, Earphones) = 3$$

$$Support(Mobile, Charger) = 2$$

$$Support(Earphones, Charger) = 2$$

After pruning:

$$L_2 = \{\{Mobile, Earphones\}\}$$

The other two itemsets are removed because their support count is less than 3.

Step 3: Generate 3-Itemset

The possible itemset is:

$$\{Mobile, Earphones, Charger\}$$

It occurs only in O4.

$$Support = 1$$

Since:

$$1 < 3$$

it is pruned.

Therefore, the algorithm terminates.

Apriori Pseudocode:

```
APRIORI(Database, MinimumSupport)

1. Scan the database.
2. Find support count of every 1-itemset.
3. Remove itemsets below MinimumSupport.
4. Store remaining itemsets as L1.
5. Set k = 2.

6. WHILE L(k-1) is not empty DO

    Generate candidate itemsets Ck
    from L(k-1).

    Prune candidates if any of their
    subsets are not frequent.

    Scan the database.

    Calculate support count of
    every remaining candidate.

    Remove candidates below
    MinimumSupport.

    Store remaining candidates as Lk.

    k = k + 1

END WHILE

7. Return all frequent itemsets.
```

Result:

The main frequent combination obtained is:

{Mobile, Earphones}

Apriori principle: If an itemset is infrequent, its supersets cannot be frequent.

Q2. FP-Growth for Library Book Borrowing Records

The dataset contains borrowing records involving **Data Science, Python, and Machine Learning**. The question specifically asks for FP-Growth without generating candidate itemsets.

Given Dataset

Borrow ID Books

B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Step 1: Calculate Frequency

Book Count

Python 4

Machine Learning 4

Data Science 3

Assume: MINIMUM SUPPORT COUNT = 3

All three books are frequent.

Step 2: Arrange Items

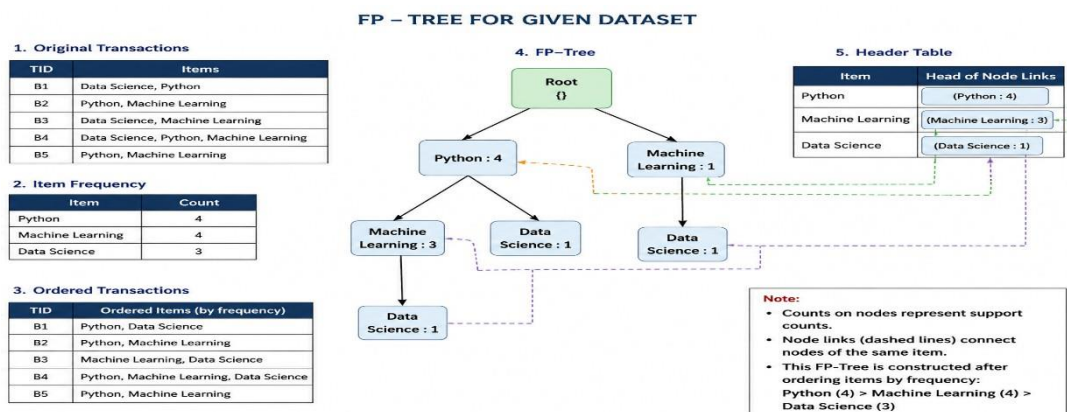
Items are arranged according to their frequency. A possible order is:

Transactions can then be reordered according to this frequency.

Step 3: Construct FP-Tree

The reordered transactions are inserted into a common tree.

A simplified representation is:



Common prefixes are stored only once, which reduces the size of the database representation.

FP-Growth Pseudocode:

```
FP-GROWTH(Database, MinimumSupport)

1. Scan the database.
2. Calculate frequency of every item.
3. Remove items below MinimumSupport.
4. Sort remaining items according to frequency.
5. Create an empty FP-Tree.
6. Insert each transaction into the FP-Tree.
7. Create a header table for frequent items.
8. Start mining from the FP-Tree.

9. FOR each item in the header table DO

    Construct its conditional pattern base.

    Construct the conditional FP-Tree.

    Recursively mine the conditional tree.

    Generate frequent patterns.

END FOR

10. Return all frequent itemsets.
```

Important Point:

FP-Growth is different from Apriori because it **does not explicitly generate candidate itemsets**.

It uses:

Database→FP Tree→Conditional Pattern Base→Conditional FP Tree→Frequent Patterns

Q3. Association Rule Generation

The given frequent service combinations are **Internet, Streaming, Cloud Storage**, along with their support counts.

Given Data

Itemset	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

**Formula
for Confidence:**

For a rule:

$$X \rightarrow Y$$

the confidence is:

$$Confidence(X \rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)} \times 100$$

Rule 1: Internet $\rightarrow$ Streaming

$$Support(Internet) = 4$$

$$Support(Internet, Streaming) = 3$$

Therefore:

$$\begin{aligned} Confidence &= \frac{3}{4} \times 100 \\ &= 75\% \end{aligned}$$

Rule 2: Streaming $\rightarrow$ Internet

$$\begin{aligned} Confidence &= \frac{3}{4} \times 100 \\ &= 75\% \end{aligned}$$

Rule 3: Streaming $\rightarrow$ Cloud Storage

$$\begin{aligned} Confidence &= \frac{2}{4} \times 100 \\ &= 50\% \end{aligned}$$

Rule 4: Cloud Storage $\rightarrow$ Streaming

$$\begin{aligned} Confidence &= \frac{2}{3} \times 100 \\ &= 66.67\% \end{aligned}$$

Rule 4: Cloud Storage $\rightarrow$ Streaming

Assume Minimum Confidence = 70%

Rule	Confidence Decision
------	---------------------

Internet $\rightarrow$ Streaming 75% Accepted

Streaming $\rightarrow$ Internet 75% Accepted

Streaming $\rightarrow$ Cloud Storage 50% Rejected

Cloud Storage $\rightarrow$ Streaming 66.67% Rejected **Pseudocode:**

```
GENERATE-RULES(FrequentItemsets, MinimumConfidence)

1. FOR each frequent itemset F DO

2.   Generate all non-empty proper
     subsets X of F.

3.   FOR each subset X DO

4.     Y = F - X

5.     Calculate:
         Confidence(X  $\rightarrow$  Y)
         = Support(F) / Support(X)

6.     IF Confidence  $\geq$  MinimumConfidence THEN
         Add X  $\rightarrow$  Y to the rule set.
       ELSE
         Reject the rule.
       END IF

7.   END FOR

8. END FOR

9. Return all accepted rules.
```

Result:

For the assumed 70% threshold:

Internet $\rightarrow$ Streaming

Streaming $\rightarrow$ Internet

are accepted.

Q4. Constraint-Based Frequent Itemset Mining

The dataset contains student course enrollments. The important constraint is:

Only itemsets containing Python are considered

This constraint is explicitly specified in the question.

Given Dataset

Student Courses

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

\

Support Counts:

Python occurs in:

✦ Upgrade

$S1, S2, S4, S5$

Therefore:

$$Support(Python) = 4$$

Python + Database occurs in:

$S1, S4$

Therefore:

$$Support(Python, Database) = 2$$

Python + AI occurs in:

$S2, S4$

Therefore:

$$Support(Python, AI) = 2$$

Python + Database + AI occurs only in $S4$:

$$Support(Python, Database, AI) = 1$$

Constraint-Based Pruning

Possible 2-itemsets:

$\{Python, Database\}$

$\{Python, AI\}$

$\{Database, AI\}$

But:

$\{Database, AI\}$

does not contain Python.

Therefore it is **pruned immediately**.

This reduces the search space.

Pseudocode:

```

CONSTRAINT-MINING(Database, MinimumSupport)

1. Generate candidate itemsets.

2. FOR each candidate C DO

    IF Python is not present in C THEN
        Prune C.
    ELSE
        Calculate support of C.

        IF Support(C) >= MinimumSupport THEN
            Add C to frequent itemsets.
        ELSE
            Prune C.
        END IF
    END IF

END FOR

3. Generate larger candidates only if
   they contain Python.

4. Repeat until no further candidates exist.

5. Return frequent itemsets.

```

Advantage:

Constraint-based mining:

- Reduces candidate itemsets.
- Reduces support calculations.
- Saves processing time.
- Focuses only on relevant patterns.

Q5. Multilevel Association Rule Mining

The dataset contains a hierarchy of hospital treatments.

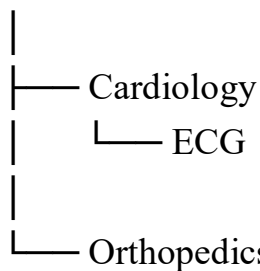
Given Dataset

Patient Treatments

P1 Healthcare, Cardiology, ECG
P2 Healthcare, Orthopedics
P3 Healthcare, Cardiology
P4 Healthcare, Cardiology, ECG
P5 Healthcare, Orthopedics

Treatment Hierarchy:

Healthcare



This allows mining at different levels.

Level 1 – General Level

Healthcare appears in all 5 transactions:

$$\begin{aligned} \text{Support}(\text{Healthcare}) &= \frac{5}{5} \times 100 \\ &= 100\% \end{aligned}$$

Level 2 – Intermediate Level

Cardiology appears in: ◆ Upgrade 🔗
 $P1, P3, P4$

Therefore:
 $\text{Support}(\text{Cardiology}) = \frac{3}{5} \times 100 = 60\%$

Orthopedics appears in:
 $P2, P5$

Therefore:
 $\text{Support}(\text{Orthopedics}) = \frac{2}{5} \times 100 = 40\%$

Level 3 – Specific Level

ECG appears in:
 $P1, P4$

Therefore:
 $\text{Support}(\text{ECG}) = \frac{2}{5} \times 100 = 40\%$

Example Rule

Consider:

$$\text{Cardiology} \rightarrow \text{ECG}$$

Cardiology occurs 3 times and both Cardiology and ECG occur together 2 times.
Therefore:

$$\text{Confidence} = \frac{2}{3} \times 100$$
$$= 66.67\%$$

Pseudocode:

MULTILEVEL-MINING(Database, Hierarchy)

1. Start from the highest level of the concept hierarchy.
2. Calculate support of items.
3. Remove items below MinimumSupport.
4. Generate frequent itemsets.
5. Generate association rules.
6. Calculate confidence.
7. Keep rules satisfying MinimumConfidence.
8. Move to the next lower level of the hierarchy.
9. Repeat Steps 2-8.
10. Stop at the lowest hierarchy level.
11. Return all interesting rules.

Importance of Concept Hierarchy

Concept hierarchies allow data to be analyzed from **general to specific**:

Healthcare → Cardiology → ECG

This helps discover:

- Broad patterns at higher levels.
- Detailed patterns at lower levels.
- Relationships between general and specific treatments.
- More meaningful patterns for decision-making.

